## Materials

Your TA will lend your group a 3D printed model of the graph $z = f(x, y)$ of some real-valued function $f$.
**Please do NOT draw on the 3D printed model and return it to your TA at the end of tutorial.**

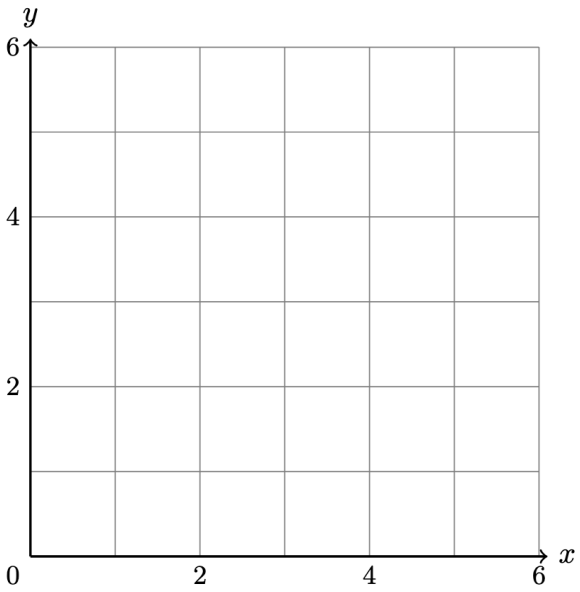
## Exercises

**Orientation:** Assume that the real-valued function $f : R \to \mathbb{R}$ has domain $R = [0, 6] \times [0, 6]$. The highest corner of your 3D model corresponds to the point $(0, 0) \in R$ and the lowest corner corresponds to the point $(6, 0) \in R$.

**Function:** Your TA will share the function formula for $f$. Record this formula below and use it for all questions on the worksheet.

$$f(x, y) = _____$$

T11.1 Consider the partition $P = (\{0, 2, 6\}, \{0, 3, 6\})$ of $R$. Record your answers to (a) and (b) using the diagram below.



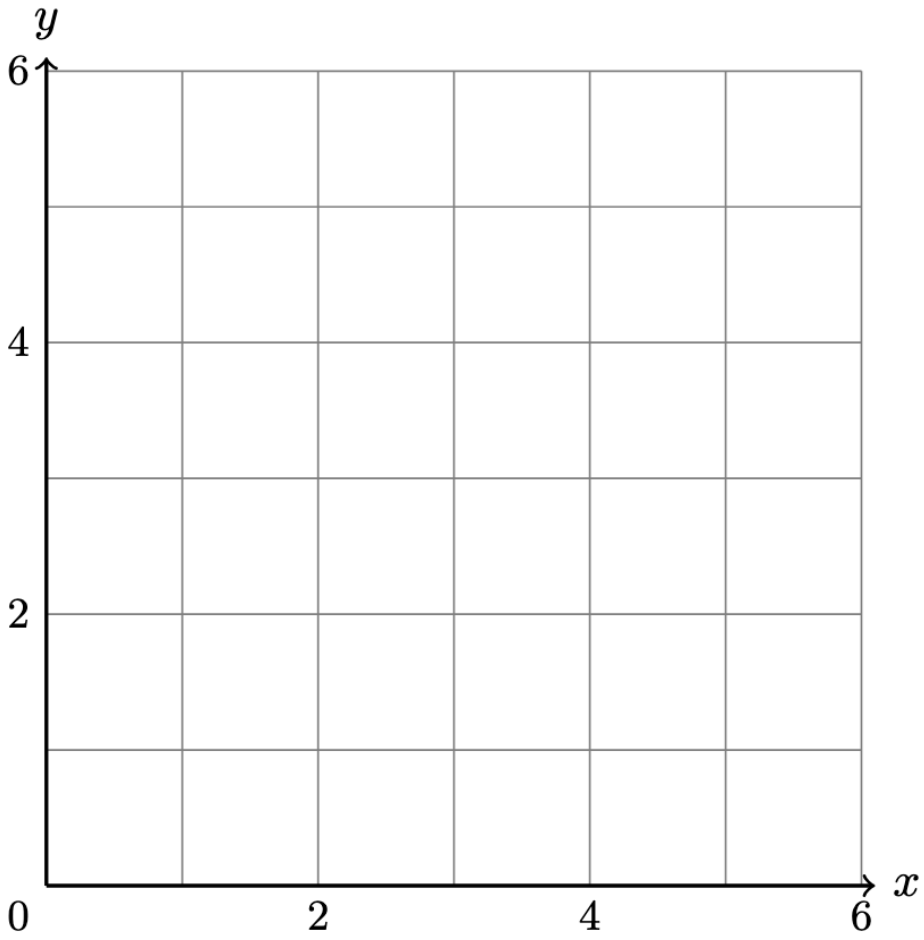

(a) Draw and label each subrectangle $R_{ij}$ of $P$ on the above diagram.

(b) Use your 3D model to guess $M_{ij} = \sup_{(x,y)\in R_{ij}} f(x, y)$ and $m_{ij} = \inf_{(x,y)\in R_{ij}} f(x, y)$ for each subrectangle $R_{ij}$ of $P$. Record the estimated values of $M_{ij}$ and $m_{ij}$ on the above diagram. Your TA will share a Desmos demo to help determine their values.

(c) Calculate the $P$-upper sum of $f$.

(d) Calculate the $P$-lower sum of $f$.

T11.2 Now consider the partition $Q = (\{0,1,2,3,4,5,6\}, \{0,1,2,3,4,5,6\})$ of $R$.




(a) Use your 3D model to guess $M_{ij} = \sup_{(x,y)\in R_{ij}} f(x,y)$ for each subrectangle $R_{ij}$ of $Q$.

Record their values on the diagram above.

(b) Calculate the $Q$-upper sum of $f$.

(c) In part (a), you secretly used that

$$\sup_{(x,y)\in R_{ij}} f(x,y) = \max_{(x,y)\in R_{ij}} f(x,y)$$

for any subrectangle $R_{ij}$ of $Q$. Why does this identity hold?

T11.3 (a) You have computed 3 out of the following 6 real numbers: $U_P(f)$, $U_Q(f)$, $L_P(f)$, $L_Q(f)$, $\overline{I_R}(f)$, $\underline{I_R}(f)$. **Without computing the other 3 unknown numbers**, write all 6 of them in ascending order, if possible.

$\leq$ $\leq$ $\leq$ $\leq$ $\leq$

Briefly explain your reasoning for these inequalities.

(b) Based on the inequalities above, is the function $f$ is integrable on $R$?

(c) Based on other theorems in your textbook, is $f$ integrable on $R$?

(d) **(Optional)** Prove $f$ is integrable on $R$ by definition using the partitions $P_N = (\{\frac{0}{N}, \frac{1}{N}, \ldots, \frac{6N}{N}\}, \{\frac{0}{N}, \frac{1}{N}, \ldots, \frac{6N}{N}\})$ for $N \in \mathbb{N}^+$. You will need the formulas

$$\sum_{k=1}^{N} k = \frac{N(N+1)}{2}, \qquad \sum_{k=1}^{N} k^2 = \frac{N(N+1)(2N+1)}{6}, \qquad \sum_{k=1}^{N} k^3 = \left(\frac{N(N+1)}{2}\right)^2.$$

Beware the proof is a long messy computation, but you will solidify your skills by at least attempting it.